HS Chemistry Resource Board Approval Talking Points

June 25, 2025

The following curriculum resource was rigorously reviewed by BBHCSD high school science teachers and administrators. This curriculum resource is available for community members to review:

AKTIV Chemistry	*From the website:* *"Aktiv Chemistry reimagines chemistry courses with an innovative approach that fosters student engagement both during and after class – in-person or online."*	https://aktiv.com/

PROCESS:

- Spring 2024 :: Planning and dialog began with teachers and administrators
- September-October 2024 :: Teachers and administrators reviewed the process and timeline
- Fall 2024 :: The committee of teachers reviewed data, goals, rubrics, and began researching materials and resources. (If applicable, the committee contacts other districts/experts to gather information on all potential products)
- October-November 2024 :: The committee ordered sample materials they researched. Options were narrowed down to 2-3 resources. (Note: a total of five other resources – including AKTIV Chemistry – were reviewed)
- December 2024 - February 2025 :: The committee convened to apply the Board Rubric and new Ohio High Quality Instructional Materials Rubric to all materials ordered for previewing
- March 2025 - April 2025 :: Additional inquiry was conducted by the curriculum department with AKTIV Chemistry
- May 1 - June 1 :: AKTIV Chemistry was available for public feedback
- June 2025 :: Board approval sought for committee recommended resources (pending a DPA)
- June 2025 :: Determine accurate roll-up enrollment numbers for 2025–2026
- July 2025 :: Order all necessary resources utilizing permanent improvement funds for FY26

In summary:

AKTIV Chemistry surpasses traditional textbooks by offering an interactive, adaptive, and accessible learning environment that actively engages students, provides immediate feedback, and supports mastery at every step. It leverages technology to make chemistry more intuitive, equitable, and effective for all learners.

A more expansive and in-depth review of AKTIV Chemistry:

1. Active, Personalized, and Immediate Learning—Not Passive Reading…..

Traditional textbooks present information linearly and require students to passively read and solve end-of-chapter problems, often without guidance or feedback. In contrast, AKTIV Chemistry transforms learning into an active, interactive process. Students engage with over 20,000 scaffolded, visual, and interactive problems—including drag-and-drop, color-coded, and multi-step questions—making abstract concepts tangible and less intimidating. The platform's real-time, targeted feedback helps students immediately identify and correct misconceptions, turning mistakes into learning opportunities and building confidence as they progress.

2. Technology-Enhanced Engagement and Accessibility…..

AKTIV Chemistry leverages technology to meet students where they are: on their devices. Its mobile-first, device-agnostic design allows students to access assignments, practice, and assessments anytime, anywhere—whether on a smartphone, tablet, or computer. This flexibility removes barriers to participation and practice, which traditional textbooks cannot match.

3. Adaptive Support, Remediation, and Enrichment…..

Unlike static textbooks, AKTIV Chemistry uses diagnostic tools to assess each student's prerequisite knowledge and automatically delivers personalized remediation or enrichment. Algorithmic problem sets can be tailored to each student's needs, ensuring that every learner is appropriately challenged and supported throughout the course.

4. Multimodal, Intuitive Learning Tools…..

AKTIV Chemistry offers unique modules that allow students to draw Lewis structures, visualize molecular geometry, balance equations, and even upload photos of their handwritten work for authentic assessment47. These features help students master skills that are difficult to convey through text alone, such as drawing organic structures or understanding spatial relationships in molecules.

5. Immediate Feedback and Multiple Attempts for Mastery…..

Students receive instant, pedagogically-intelligent feedback with every attempt, helping them learn from their errors and try again—something a textbook cannot provide. Teachers can allow multiple attempts, track student progress, and provide extended time accommodations, supporting mastery learning and accessibility for all students.

6. Teacher Flexibility and Customization…..

AKTIV Chemistry empowers teachers to select problems at various levels, create their own questions, and integrate content from primary resources or online textbooks like OpenStax or Substack. This flexibility ensures alignment with course objectives and allows for continuous improvement and adaptation to student needs.

7. Proven Outcomes and Student Endorsement…..

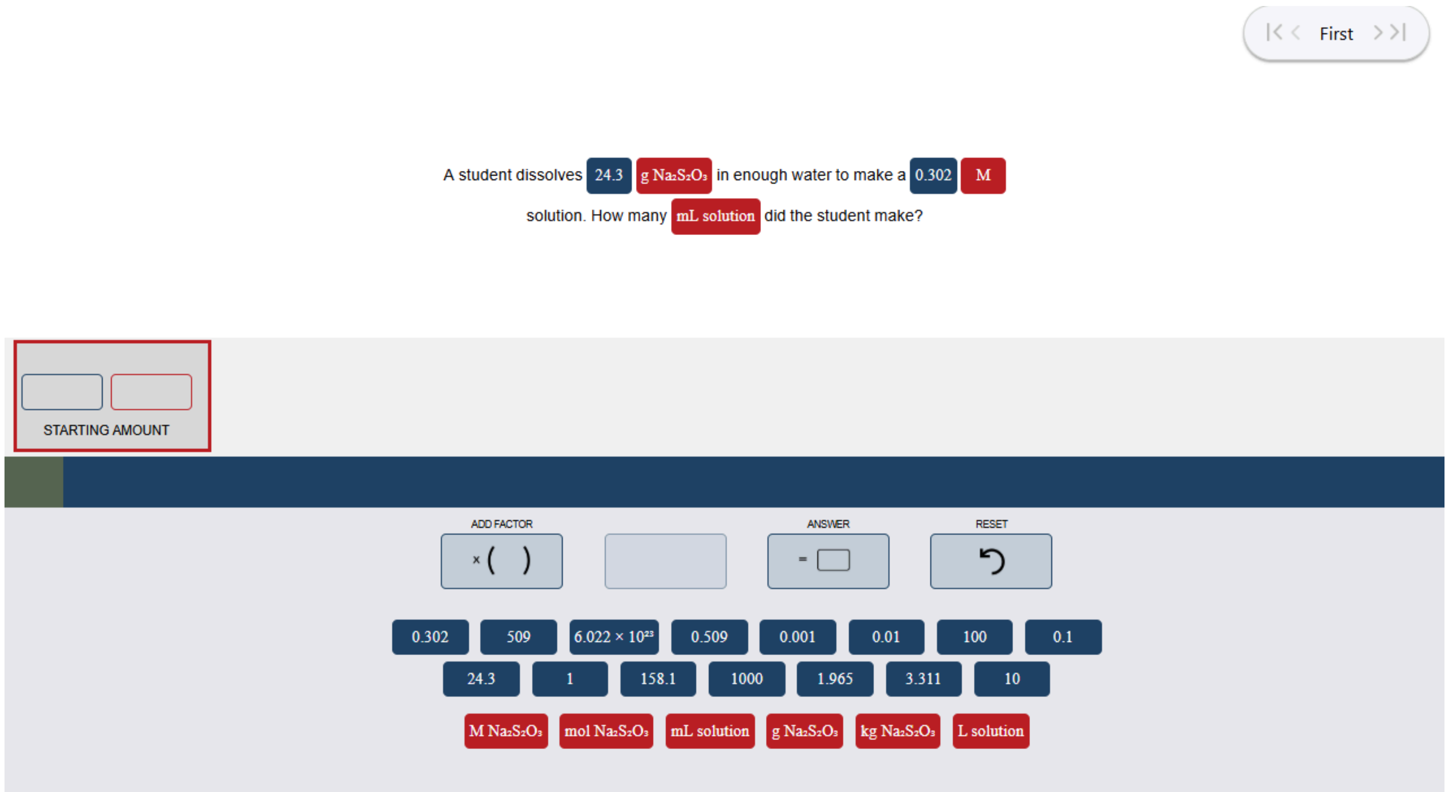
Institutions adopting AKTIV Chemistry have reported lower dropout rates, higher engagement, and improved exam preparation compared to traditional methods. According to surveys, a majority of students believe AKTIV Chemistry helped them improve their grades and recommend its continued use.

Summary Table: AKTIV Chemistry vs. Traditional Textbooks

Feature	AKTIV Chemistry
Interactivity	High (drag-and-drop, drawing, color-coded)
Feedback	Immediate, targeted, pedagogical
Accessibility	Mobile, device-agnostic, online
Adaptivity	Personalized remediation and enrichment
Engagement	Intuitive, multimodal
Assessment	Auto-graded, multi-attempt, authentic uploads
Teacher Customization	High (custom questions, levels, integration)
Student Endorsement	High (improved grades, recommend use)

Examples: Aktiv problems with student submissions:

Drag and drop blank: This is what the student initially sees; the next screenshot is completed by the student in one submission:



First

A student dissolves 24.3 g $Na_2S_2O_3$ in enough water to make a 0.302 M solution. How many mL solution did the student make?

$$24.3\ \cancel{g\ Na_2S_2O_3} \times \left(\frac{1\ \cancel{mol\ Na_2S_2O_3}}{158.1\ \cancel{g\ Na_2S_2O_3}}\right) \times \left(\frac{1\ \cancel{L\ solution}}{0.302\ \cancel{mol\ Na_2S_2O_3}}\right) \times \left(\frac{1000\ mL\ solution}{1\ \cancel{L\ solution}}\right) = 509\ mL\ solution$$

STARTING AMOUNT

$$24.3\ g\ Na_2S_2O_3 \times \frac{1\ mol\ Na_2S_2O_3}{158.1\ g\ Na_2S_2O_3} \times \frac{1\ L\ solution}{0.302\ mol\ Na_2S_2O_3} \times \frac{1000\ mL\ solution}{1\ L\ solution} = 509\ mL\ solution$$

Correct!

The correct answer is 509 mL solution.

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

This problem was attempted by the student three times. Below is the blank problem and the student's three submissions.

Glycerol ($C_3H_8O_3$) is a byproduct of soapmaking. What is the molality of a solution made by dissolving 155 g $C_3H_8O_3$ per L water ? The density of water is 1 kg water/L .

STARTING AMOUNT

ADD FACTOR × ( )

ANSWER

RESET

1680 | 6.022×10^{23} | 1.68 | 0.594 | 594 | 0.001 | 155 | 16.8 | 1 | 1000 | 15.5 | 59.4 | 92.09

mol $C_3H_8O_3$ | m $C_3H_8O_3$ | kg water/L | mL water | kg water | g $C_3H_8O_3$/mol | g $C_3H_8O_3$/L water | g $C_3H_8O_3$ | L water

Glycerol ($C_3H_8O_3$) is a byproduct of soapmaking. What is the molality of a solution made by dissolving 155 g $C_3H_8O_3$ per L water ? The density of water is 1 kg water/L .

155 g $C_3H_8O_3$ × (1 L water / 92.09 kg water) = 1.68 kg water / L water

STARTING AMOUNT

$$155\ g\ C_3H_8O_3 \times \frac{1\ L\ water}{92.09\ kg\ water} = \frac{1.68\ kg\ water}{L\ water}$$

Incorrect, 2 attempts remaining

The correct answer is 1.68 mol $C_3H_8O_3$ / kg water.

Your ratio between L water and kg water in your first conversion factor is incorrect. Check the problem statement and then try to determine the correct values for the factor.

Glycerol ($C_3H_8O_3$) is a byproduct of soapmaking. What is the molality of a solution made by dissolving 155 g $C_3H_8O_3$ per L water ? The density of water is 1 kg water/L .

155 g $C_3H_8O_3$ × (1 L water / 92.09 kg water) = 1.68 g $C_3H_8O_3$ / L water

STARTING AMOUNT

$$155\ g\ C_3H_8O_3 \times \frac{1\ L\ water}{92.09\ kg\ water} = \frac{1.68\ g\ C_3H_8O_3}{L\ water}$$

Incorrect, 1 attempt remaining

The correct answer is 1.68 mol $C_3H_8O_3$ / kg water.

Your ratio between L water and kg water in your first conversion factor is incorrect. Check the problem statement and then try to determine the correct values for the factor.

Glycerol ($C_3H_8O_3$) is a byproduct of soapmaking. What is the molality of a solution made by dissolving 155 g $C_3H_8O_3$ per L water ? The density of water is 1 kg water/L .

155 g $C_3H_8O_3$ × (1 L water / 92.09 kg water) = 1.68 mol $C_3H_8O_3$ / kg water

STARTING AMOUNT

$$155 \text{ g } C_3H_8O_3 \times \frac{1 \text{ L water}}{92.09 \text{ kg water}} = \frac{1.68 \text{ mol } C_3H_8O_3}{\text{kg water}}$$

Incorrect, 0 attempts remaining

The correct answer is 1.68 mol $C_3H_8O_3$ / kg water.

Your ratio between L water and kg water in your first conversion factor is incorrect. Check the problem statement and then try to determine the correct values for the factor.